

# The Base-Rate Trap

Day 2 · The Eval Gap · 50 min studio (the math is the lecture) · pods of 3-5

**Objective.** Build the confusion matrix by hand from a base rate and a judge's TPR/FPR, compute the PPV and the miss count, then work out the system's **effective sensitivity** — and decide what “zero flags last month” really means.

**ClearList, by the numbers.** Meridian's AML triage agent sees **50,000 alerts / month**. The true base rate of suspicious activity is **2%**. The vendor reports the LLM judge runs at **TPR = 90%** and **FPR = 5%**. The agent auto-clears everything it doesn't escalate. Sounds like a strong judge. Let's check what it actually does.

**VOCABULARY (POSITIVE = “SUSPICIOUS / ESCALATE”)**

- **TPR** (true-positive rate; recall, sensitivity) — of truly suspicious alerts, the share the judge escalates.
- **FPR** (false-positive rate) — of non-suspicious alerts, the share the judge wrongly escalates.
- **PPV** (positive predictive value; precision) — of everything the judge escalates, the share that is truly suspicious.
- **FNR** (false-negative rate; miss rate) =  $1 - \text{TPR}$  — of truly suspicious alerts, the share the judge wrongly clears.

$\text{PPV} = \text{TP} / (\text{TP} + \text{FP})$  ·  $\text{FNR} = \text{FN} / (\text{FN} + \text{TP})$

## Step 1 · Build the matrix (≈10 min)

Fill in the four cells for 50,000 alerts. Start from the base rate, then split each column by TPR / FPR.

|                 | TRULY SUSPICIOUS | NOT SUSPICIOUS | ROW TOTAL |
|-----------------|------------------|----------------|-----------|
| Judge: ESCALATE | TP =             | FP =           |           |
| Judge: CLEAR    | FN =             | TN =           |           |
| Column total    |                  |                | 50,000    |

## Step 2 · The numbers that matter (≈6 min)

PPV =  $\text{TP} / (\text{TP} + \text{FP}) =$  \_\_\_\_\_

FNR =  $\text{FN} / (\text{FN} + \text{TP}) =$  \_\_\_\_\_

Truly-suspicious alerts auto-cleared each month (FN) = \_\_\_\_\_

Of everything escalated, what fraction is a false alarm? \_\_\_\_\_

Before Step 3: a **90% TPR** judge sounds excellent. Does your PPV agree? What surprised you?

## Step 3 · Effective sensitivity (≈8 min)

The escalated pile goes to human investigators. Under the current flood, assume investigators catch **70%** of the real cases that reach their queue.

Effective sensitivity = judge TPR  $\times$  reviewer hit rate =  $0.90 \times 0.70 =$  \_\_\_\_\_

So the system as a whole catches \_\_\_\_\_% of truly suspicious activity – and misses \_\_\_\_\_%.

How many false alarms are flooding the investigator queue each month? \_\_\_\_\_

Now the twist: does flooding the queue with false alarms make that 70% reviewer hit rate go *up* or *down*?

#### Step 4 · “Zero flags last month” ( $\approx$ debrief)

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Management reports that **zero** confirmed misses came out of the auto-cleared pile last month. The auto-cleared pile is never sampled. **Write one sentence:** is “zero confirmed misses” evidence that the agent missed nothing?

**Illustrative for the exercise only.** The numbers are ClearList’s stated figures; real judge performance must be measured on a labeled gold set, not assumed. Nothing here is legal or regulatory advice.